Introduction to Machine Learning, Spring 2025
Homework 1

(Due March 7, 2025 at 11:59pm (CST))

March 5, 2025

1. Please write your solutions in English.
2. Submit your solutions to the course Gradescope.
3. If you want to submit a handwritten version, scan it clearly.
4. Late homeworks submitted within 3 days of the due date will be marked down 25% each day cumulatively Homeworks submitted more than 3 days after the due date will not be accepted unless there is a valid reason, such as a medical or family emergency.
5. You are required to follow ShanghaiTech's academic honesty policies. You are allowed to discuss problems with other students, but you must write up your solutions by yourselves. You are not allowed to copy materials from other students or from online or published resources. Violating academic honesty can result in serious penalties.

1. [15 points] [Math review(Linear Algebra)] Rank properties. Suppose a matrix $A \in \mathbb{R}^{m \times n}$, prove:

   (a) $\mathbf{rank}(A) \leq \min(m, n)$ [4 points] (Hint: You just need to consider the foundamental transformation of the matrix. And how **rank** is defined.)

   (b) $\mathbf{rank}(AB) \leq \min(\mathbf{rank}(A), \mathbf{rank}(B))$. [4 points]

   (c) $\mathbf{rank}(A^\top A) = \mathbf{rank}(A)$. [4 points] (Hint: consider the identity between **rank** and **nullity**)

   (d) What does the rank of a matrix essentially refer to? (Hint: consider the correspondence to the singularvalues.) [3 points]

Solution

(a) The rank of a matrix A is defined as the number of linearly independent rows or columns. When performing fundamental transformations (row or column operations), the rank of the matrix remains unchanged. Since A is an $m \times n$ matrix, it has m rows and n columns. The maximum number of linearly independent rows cannot exceed m, and the maximum number of linearly independent columns cannot exceed n. Therefore, $\mathbf{rank}(A) \leq \min(m, n)$.

(b) Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. Then $AB \in \mathbb{R}^{m \times p}$. The column space of AB, denoted as $\mathcal{C}(AB)$, is a subspace of the column space of A, i.e., $\mathcal{C}(AB) \subseteq \mathcal{C}(A)$. Therefore, $\mathbf{rank}(AB) = \dim(\mathcal{C}(AB)) \leq \dim(\mathcal{C}(A)) = \mathbf{rank}(A)$. Similarly, the row space of AB is a subspace of the row space of B. Thus, $\mathbf{rank}(AB) \leq \mathbf{rank}(B)$. Combining these two inequalities, we have $\mathbf{rank}(AB) \leq \min(\mathbf{rank}(A), \mathbf{rank}(B))$.

(c) We want to show that $\mathbf{rank}(A^\top A) = \mathbf{rank}(A)$. First, note that $A^\top A \in \mathbb{R}^{n \times n}$. We will use the fact that $\mathbf{rank}(A) + \mathbf{nullity}(A) = n$, where $\mathbf{nullity}(A)$ is the dimension of the null space of A. We want to show that $\mathbf{nullity}(A^\top A) = \mathbf{nullity}(A)$. If $x \in \mathcal{N}(A)$, then $Ax = 0$, so $A^\top Ax = A^\top 0 = 0$, which means $x \in \mathcal{N}(A^\top A)$. Thus, $\mathcal{N}(A) \subseteq \mathcal{N}(A^\top A)$. Conversely, if $x \in \mathcal{N}(A^\top A)$, then $A^\top Ax = 0$. Multiplying by $x^\top$ on the left, we get $x^\top A^\top Ax = (Ax)^\top (Ax) = \|Ax\|^2 = 0$. This implies that $Ax = 0$, so $x \in \mathcal{N}(A)$. Thus, $\mathcal{N}(A^\top A) \subseteq \mathcal{N}(A)$. Therefore, $\mathcal{N}(A^\top A) = \mathcal{N}(A)$, and $\mathbf{nullity}(A^\top A) = \mathbf{nullity}(A)$. Now, $\mathbf{rank}(A^\top A) = n - \mathbf{nullity}(A^\top A) = n - \mathbf{nullity}(A) = \mathbf{rank}(A)$.

(d) The rank of a matrix essentially refers to the number of non-zero singular values of the matrix. The singular value decomposition (SVD) of a matrix A is given by $A = U\Sigma V^\top$, where Σ is a diagonal matrix containing the singular values of A. The number of non-zero singular values corresponds to the number of linearly independent rows or columns of A, which is the rank of A.

2. [15 points] [Math review(Linear Algebra)] Eigenvalue Decomposition(EVD) / Spectral Decomposition. Find the geometric and algebraic multiplicity of each eigenvalue of the following matrix A, and determine whether A is diagonalizable. If A is diagonalizable, then find a matrix P that diagonalizes A, and find the diagonal matrix Λ such that $\Lambda = P^{-1}AP$. Furthermore, if A could be orthogonally diagonalized, write Λ as $\Lambda = P^{\top}AP$.

(a) $A = \begin{bmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{bmatrix}$. [5 points]

(b) $A = \begin{bmatrix} 5 & 0 & 0 \\ 1 & 5 & 0 \\ 0 & 1 & 5 \end{bmatrix}$. [5 points]

(c) $A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{bmatrix}$. [5 points]

Solution

(a) $A = \begin{bmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{bmatrix}$.

for A we have the characteristic polynomial $p(\lambda) = \det(A - \lambda I) = 0$.

$$\begin{aligned} p(\lambda) &= \begin{vmatrix} -1-\lambda & 4 & -2 \\ -3 & 4-\lambda & 0 \\ -3 & 1 & 3-\lambda \end{vmatrix} \\ &= -\lambda^3 + 6\lambda^2 - 11\lambda + 6 \\ &= -(\lambda - 1)(\lambda - 2)(\lambda - 3) \end{aligned}$$

Thus, the eigenvalues are $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3$.

For $\lambda_1 = 1$, we have $A - \lambda_1 I = \begin{bmatrix} -2 & 4 & -2 \\ -3 & 3 & 0 \\ -3 & 1 & 2 \end{bmatrix}$.

Reduced row echelon form of $A - \lambda_1 I$ is $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

The eigenvector corresponding to $\lambda_1 = 1$ is $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. The geometric multiplicity of $\lambda_1 = 1$ is 1 and the algebraic multiplicity is 1.

For $\lambda_2 = 2$, we have $A - \lambda_2 I = \begin{bmatrix} -3 & 4 & -2 \\ -3 & 2 & 0 \\ -3 & 1 & 1 \end{bmatrix}$.

Reduced row echelon form of $A - \lambda_2 I$ is $\begin{bmatrix} 1 & 0 & -\frac{2}{3} \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$.

The eigenvector corresponding to $\lambda_2 = 2$ is $\begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}$. The geometric multiplicity of $\lambda_2 = 2$ is 1 and the algebraic multiplicity is 1.

For $\lambda_3 = 3$, we have $A - \lambda_3 I = \begin{bmatrix} -4 & 4 & -2 \\ -3 & 1 & 0 \\ -3 & 1 & 0 \end{bmatrix}$.

Reduced row echelon form of $A - \lambda_3 I$ is $\begin{bmatrix} 1 & 0 & -\frac{1}{4} \\ 0 & 1 & -\frac{3}{4} \\ 0 & 0 & 0 \end{bmatrix}$.

The eigenvector corresponding to $\lambda_3 = 3$ is $\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$. The geometric multiplicity of $\lambda_3 = 3$ is 1 and the algebraic multiplicity is 1.

Since the algebraic multiplicity of each eigenvalue is equal to the geometric multiplicity of each eigenvalue,

A is diagonalizable.
We can obtain the matrix P by taking the eigenvectors of A as columns of P.

$$P = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

The diagonal matrix Λ is given by the eigenvalues of A on the diagonal.

$$\Lambda = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

If A could be orthogonally diagonalized, then A must be symmetric. Since A is not symmetric, A cannot be orthogonally diagonalized.

(b) $A = \begin{bmatrix} 5 & 0 & 0 \\ 1 & 5 & 0 \\ 0 & 1 & 5 \end{bmatrix}$.
Because A is a diagonal matrix, the eigenvalues are the diagonal entries of A. Thus, the eigenvalues are $\lambda_1 = 5, \lambda_2 = 5, \lambda_3 = 5$.
For $\lambda_1 = 5$, we have $A - \lambda_1 I = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$.
Reduced row echelon form of $A - \lambda_1 I$ is $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.
The eigenvector corresponding to $\lambda_1 = 5$ is $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. The geometric multiplicity of $\lambda_1 = 5$ is 1 and the algebraic multiplicity is 3.
Samely, the geometric multiplicity of $\lambda_2 = 5$ and $\lambda_3 = 5$ is 1 and the algebraic multiplicity is 3.
As we can see, here the algebraic multiplicity of each eigenvalue is not equal to the geometric multiplicity of each eigenvalue. Thus, A is not diagonalizable.

(c) $A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{bmatrix}$.
for A we have the characteristic polynomial $p(\lambda) = \det(A - \lambda I) = 0$.

$$\begin{aligned} p(\lambda) &= \begin{vmatrix} 4-\lambda & 2 & 2 \\ 2 & 4-\lambda & 2 \\ 2 & 2 & 4-\lambda \end{vmatrix} \\ &= -\lambda^3 + 12\lambda^2 - 36\lambda + 32 \\ &= -(\lambda - 2)^2(\lambda - 8) \end{aligned}$$

Thus, the eigenvalues are $\lambda_1 = 2, \lambda_2 = 2, \lambda_3 = 8$.
For $\lambda_1 = 2$, we have $A - \lambda_1 I = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}$.
Reduced row echelon form of $A - \lambda_1 I$ is $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.
The eigenvectors corresponding to 2 are $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$. The geometric multiplicity of $\lambda_1 = 2$ is 2 and the algebraic multiplicity is 2.
Samely, the geometric multiplicity of $\lambda_2 = 2$ is 2 and the algebraic multiplicity is 2.
For $\lambda_3 = 8$, we have $A - \lambda_3 I = \begin{bmatrix} -4 & 2 & 2 \\ 2 & -4 & 2 \\ 2 & 2 & -4 \end{bmatrix}$.

Reduced row echelon form of $A - \lambda_3 I$ is $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$.

The eigenvector corresponding to $\lambda_3 = 8$ is $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. The geometric multiplicity of $\lambda_3 = 8$ is 1 and the algebraic multiplicity is 1.

Since the algebraic multiplicity of each eigenvalue is equal to the geometric multiplicity of each eigenvalue, A is diagonalizable.

We can obtain the matrix P by taking the eigenvectors of A as columns of P.

$$P = \begin{bmatrix} -1 & -1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

The diagonal matrix Λ is given by the eigenvalues of A on the diagonal.

$$\Lambda = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

A is symmetric, so A could be orthogonally diagonalized. Thus, we can write $\Lambda = P^\top AP$.

To obtain the orthogonal matrix P, we can use the Gram-Schmidt process.

Let $v_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$, and $v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

The orthogonal basis for v_1 is $u_1 = \frac{v_1}{||v_1||} = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$.

The orthogonal basis for v_2 is $u_2 = \frac{v_2 - (v_2 \cdot u_1)u_1}{||v_2 - (v_2 \cdot u_1)u_1||} = \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \end{bmatrix}$.

The orthogonal basis for v_3 is $u_3 = \frac{v_3 - (v_3 \cdot u_1)u_1 - (v_3 \cdot u_2)u_2}{||v_3 - (v_3 \cdot u_1)u_1 - (v_3 \cdot u_2)u_2||} = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}$.

Thus, the orthogonal matrix P is given by

$$P = \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

the diagonal matrix Λ is given by

$$\Lambda = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

3. [15 points] [Math review(Calculus)] Differential calculus of functions.

   (a) Suppose $f : \mathbb{R} \to \mathbb{R}$ is a second order differentiable function. Write down the second order Taylor expansion of $f(x)$ around x_0 with Peano's form of the remainder. [3 points]

   (b) Suppose $f : \mathbb{R} \to \mathbb{R}$ is a second order differentiable function. Write down the first order and second order's mean value theorem for $f(x)$ related to point x and $x + d$. [4 points]

   (c) Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is a second order differentiable function. Write down the second order Taylor expansion of $f(\mathbf{x})$ around $\mathbf{x}_0$ with Peano's form of the remainder. [4 points] (Hint: the first order derivative is the gradient ∇f, the second order derivative is the Hessian $\nabla^2 f$.)

   (d) Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is a second order differentiable function. Write down the first order and second order's mean value theorem for $f(\mathbf{x})$ related to point $\mathbf{x}$ and $\mathbf{x} + \mathbf{d}$. [4 points]

Solution

(a) The second order Taylor expansion of $f(x)$ around x_0 with Peano's form of the remainder is given by

$$\begin{aligned} f(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + R_2(x) \\ &= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + o((x - x_0)^2) \end{aligned}$$

where $R_2(x) = o((x - x_0)^2)$.

(b) The first order and second order's mean value theorem for $f(x)$ related to point x and $x+d$ are given by

- First order mean value theorem: There exists a point c between x and $x + d$ such that

$$f(x + d) - f(x) = f'(c)d$$

- Second order mean value theorem: There exists a point c between x and $x + d$ such that

$$f(x + d) - f(x) = f'(x)d + \frac{f''(c)}{2}d^2$$

(c) The second order Taylor expansion of $f(\mathbf{x})$ around $\mathbf{x}_0$ with Peano's form of the remainder is given by

$$\begin{aligned} f(\mathbf{x}) &= f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^T(\mathbf{x} - \mathbf{x}_0) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^T\nabla^2 f(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0) + R_2(\mathbf{x}) \\ &= f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^T(\mathbf{x} - \mathbf{x}_0) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^T\nabla^2 f(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0) + o(\|\mathbf{x} - \mathbf{x}_0\|^2) \end{aligned}$$

where $R_2(\mathbf{x}) = o(\|\mathbf{x} - \mathbf{x}_0\|^2)$.

(d) The first order and second order's mean value theorem for $f(\mathbf{x})$ related to point $\mathbf{x}$ and $\mathbf{x} + \mathbf{d}$ are given by

- First order mean value theorem: There exists a point $\mathbf{c}$ between $\mathbf{x}$ and $\mathbf{x} + \mathbf{d}$ such that

$$f(\mathbf{x} + \mathbf{d}) - f(\mathbf{x}) = \nabla f(\mathbf{c})^T\mathbf{d}$$

- Second order mean value theorem: There exists a point $\mathbf{c}$ between $\mathbf{x}$ and $\mathbf{x} + \mathbf{d}$ such that

$$f(\mathbf{x} + \mathbf{d}) - f(\mathbf{x}) = \nabla f(\mathbf{x})^T\mathbf{d} + \frac{1}{2}\mathbf{d}^T\nabla^2 f(\mathbf{c})\mathbf{d}$$

4. [15 points] [Math review(Probability and Statistics)] Suppose we have two random variables, A and B, where A takes on values a_1, a_2 and B takes on values b_1, b_2. Let $P(A = a_1) = 0.5$ and $P(B = b_1) = 0.5$.

   (a) Suppose a_1 and b_2 are disjoint (mutually exclusive). [3 points]
   1. What is $P(a_1, b_2)$?
   2. What is $P(a_1, b_1)$?
   3. What is $P(a_1 \mid b_2)$?

   (b) Suppose instead that A, B are independent. [3 points]
   1. What is $P(a_1, b_2)$?
   2. What is $P(a_1, b_1)$?
   3. What is $P(a_1 \mid b_2)$?

   (c) A student is looking at her activity tracker (Fitbit/Apple Watch) data and she notices that she seems to sleep better on days that she exercises. They observe the following:

Exercise	Good Sleep	Probability
yes	yes	0.3
yes	no	0.2
no	no	0.4
no	yes	0.1

   1. What is the $P(\text{GoodSleep} = \text{yes} \mid \text{Exercise} = \text{yes})$?
   2. Why doesn't $P(\text{GoodSleep} = \text{yes}, \text{Exercise} = \text{yes}) = P(\text{GoodSleep} = \text{yes}) \cdot P(\text{Exercise} = \text{yes})$?
   3. The student merges her activity tracker data with her food logs and finds that the $P(\text{Eatwell} = \text{yes} \mid \text{Exercise} = \text{yes}, \text{GoodSleep} = \text{yes})$ is 0.25. What is the probability of all three happening on the same day? [3 points]

   (d) What is the expectation of X where X is a single roll of a fair 6-sided die ($S = \{1, 2, 3, 4, 5, 6\}$)? What is the variance of X? [3 points]

   (e) Imagine that we had a new die where the sides were $S = \{3, 4, 5, 6, 7, 8\}$. How do the expectation and the variance compare to our original dice? [3 points]

Solution

(a) Suppose a_1 and b_2 are disjoint (mutually exclusive).
- 1. $P(a_1, b_2) = 0$
- 2. $P(a_1, b_1) = P(a_1) = 0.5$
- 3. $P(a_1 \mid b_2) = \frac{P(a_1, b_2)}{P(b_2)} = 0$

(b) Suppose instead that A, B are independent.
- 1. $P(a_1, b_2) = P(a_1)\,P(b_2) = 0.5 \times 0.5 = 0.25$
- 2. $P(a_1, b_1) = P(a_1)\,P(b_1) = 0.5 \times 0.5 = 0.25$
- 3. $P(a_1 \mid b_2) = \frac{P(a_1, b_2)}{P(b_2)} = \frac{0.25}{0.5} = 0.5$

(c)
- 1. $P(\text{GoodSleep} = \text{yes} \mid \text{Exercise} = \text{yes}) = \frac{P(\text{GoodSleep=yes, Exercise=yes})}{P(\text{Exercise=yes})} = \frac{0.3}{0.3+0.2} = \frac{0.3}{0.5} = 0.6$
- 2. $P(\text{GoodSleep} = \text{yes}, \text{Exercise} = \text{yes}) = P(\text{GoodSleep} = \text{yes}) \cdot P(\text{Exercise} = \text{yes})$ is not true because the events are not independent.
- 3. $P(\text{Eatwell} = \text{yes}, \text{Exercise} = \text{yes}, \text{GoodSleep} = \text{yes}) = P(\text{Eatwell} = \text{yes} \mid \text{Exercise} = \text{yes}, \text{GoodSleep} = \text{yes}) \cdot P(\text{Exercise} = \text{yes}, \text{GoodSleep} = \text{yes}) = 0.25 \cdot 0.3 = 0.075$

(d) The expectation of X where X is a single roll of a fair 6-sided die ($S = \{1, 2, 3, 4, 5, 6\}$) is given by

$$E[X] = \sum_{x \in S} xP(X = x) = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = 3.5$$

The variance of X is given by

$$\text{Var}(X) = E[X^2] - E[X]^2 = \frac{1}{6}\left(1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2\right) - 3.5^2 = \frac{35}{12} \approx 2.9167$$

(e) The expectation of X where X is a single roll of a fair 6-sided die ($S = \{3, 4, 5, 6, 7, 8\}$) is given by

$$E[X] = \sum_{x \in S} xP(X = x) = \frac{1}{6}(3 + 4 + 5 + 6 + 7 + 8) = 5.5$$

The variance of X is given by

$$\text{Var}(X) = E[X^2] - E[X]^2 = \frac{1}{6}\left(3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2\right) - 5.5^2 = \frac{35}{12} \approx 2.9167$$

5. [20 points] [Math review(Information Theory)] Example of joint entropy. Let $p(x, y)$ be given by

X \ Y	0	1
0	$\frac{1}{3}$	$\frac{1}{3}$
1	0	$\frac{1}{3}$

Find:

(a) $H(X), H(Y)$. [4 points]
(b) $H(X \mid Y), H(Y \mid X)$. [4 points]
(c) $H(X, Y)$. [3 points]
(d) $H(Y) - H(Y \mid X)$. [1 points]
(e) $I(X; Y)$. [1 points]
(f) Draw a Venn diagram for the quantities in parts (a) through (e). [3 points]
(g) When is the mutual information $I(X; Y) = 0$? [4 points]

Solution

(a)

$$p(x) = \begin{cases} \frac{2}{3} & x = 0 \\ \frac{1}{3} & x = 1 \end{cases}$$
$$p(y) = \begin{cases} \frac{1}{3} & y = 0 \\ \frac{2}{3} & y = 1 \end{cases}$$

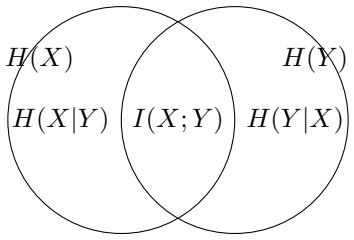
The entropy of X and Y are given by

$$H(X) = -\sum_x p(x) \log_2 p(x) = \frac{2}{3} \log_2 \frac{3}{2} + \frac{1}{3} \log_2 3$$
$$H(Y) = -\sum_y p(y) \log_2 p(y) = \frac{1}{3} \log_2 3 + \frac{2}{3} \log_2 \frac{3}{2}$$

(b) $H(X \mid Y) = H(X \mid Y = 0)p(Y = 0) + H(X \mid Y = 1)p(Y = 1) = 0 + \frac{2}{3} \cdot (\frac{1}{2} \log_2 2 + \frac{1}{2} \log_2 2) = \frac{2}{3}$

$H(Y \mid X) = H(Y \mid X = 0)p(X = 0) + H(Y \mid X = 1)p(X = 1) = \frac{2}{3} \cdot (\frac{1}{2} \log_2 2 + \frac{1}{2} \log_2 2) + 0 = \frac{2}{3}$

(c) $H(X, Y) = -\sum_x \sum_y p(x, y) \log_2 p(x, y) = -\frac{1}{3} \log_2 \frac{1}{3} - \frac{1}{3} \log_2 \frac{1}{3} - \frac{1}{3} \log_2 \frac{1}{3} = \log_2 3$
(d) $H(Y) - H(Y \mid X) = \frac{1}{3} \log_2 3 + \frac{2}{3} \log_2 \frac{3}{2} - \frac{2}{3}$
(e) $I(X; Y) = H(Y) - H(Y \mid X) = \frac{1}{3} \log_2 3 + \frac{2}{3} \log_2 \frac{3}{2} - \frac{2}{3}$



$H(X, Y)$ = entire area of both circles

(f)

(g) The mutual information $I(X;Y) = 0$ when X and Y are independent. In this case, $p(x,y) = p(x)p(y)$.

$$\begin{aligned} I(X;Y) &= \sum_{x,y} p(x,y) \log_2 \frac{p(x,y)}{p(x)p(y)} \\ &= \sum_{x,y} p(x,y) \log_2 \frac{p(x)p(y)}{p(x)p(y)} \\ &= \sum_{x,y} p(x,y) \log_2 1 \\ &= \sum_{x,y} p(x,y) \cdot 0 \\ &= 0 \end{aligned}$$